CS 4391 Introduction to Computer Vision
Homework 2

Professor Yu Xiang

February 8, 2024

Download the homework2_programming.zip file from eLearning, Assignments, Homework 2. Finish the following programming problems and submit your scripts to eLearning. You can zip all the data and files for submission. Our TA will run your scripts to verify them.

Install the Python packages needed by

- pip install -r requirement.txt

Here are some useful resources:

- Python basics `https://pythonbasics.org/`
- Numpy `https://numpy.org/doc/stable/user/basics.html`
- OpenCV `https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html`

Problem 1

(4 points) Edge Detection.

Implement the edge_detection() function in edge_detection.py. The script implements the edge detection algorithm. Follow the steps in the script to implement it.

After your implementation, run the edge_detection.py in Python to verify it. Figure 1 shows an example of running the script.

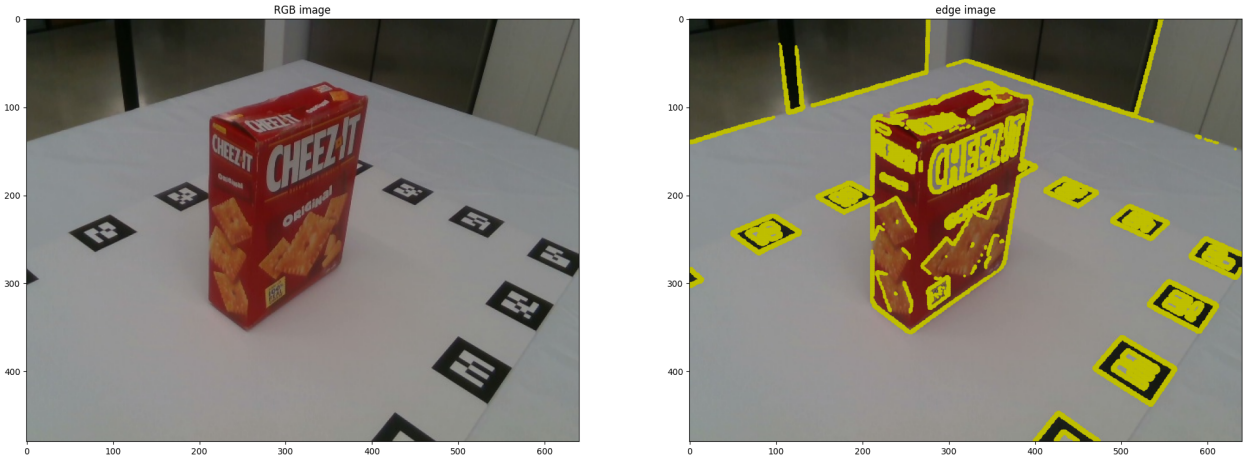


Figure 1: (Left) An input image. (Right) Edge detection with my implementation.

Problem 2

(6 points) Harris Corner Detector.

Implement the harris_corner() function and the non_maximum_suppression() function in harris_corner.py. The script implements the Harris corner detection algorithm. Follow the steps in the script to implement it.

After your implementation, run the harris_corner.py in Python to verify it. Figure 2 shows an example of running the script.

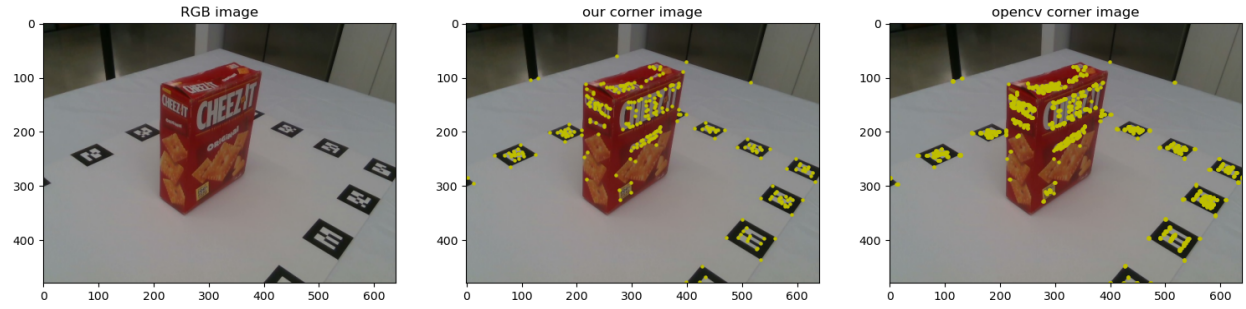


Figure 2: (Left) An input image. (Middle) Harris corner detection with our implementation. (Right) Harris corner detection using the provided function in OpenCV. The yellow dots indicate the detected corners.